

Date: _____

December 7, 2026 · 5:00 – 7:30 PM · Kai Santos

NOTES

01 Corrosion Fundamentals & Types

02 Protection, Material Selection & Inspection

03 Course Wrap-up

SLIDE 2 *Class Agenda*

EXAMET-5-R-600-HD • 50-500x

Specimens to Check Out

- No new specimen — review notes from Modules 1-14

How to check out the kit:

- 1 See instructor after class today
- 2 Sign out the specimen tray
- 3 Use the EXAMET-5 at your convenience
- 4 Return kit before leaving class
- 5 Submit completed module with your portfolio

Portfolio Handout — Module 15

- SLIDE 3** *Metallography Lab — Module 15 EXAMET-5-R-600-HD · 50–500×*

MECH 191 — Course & Unit Roadmap

Unit 1	Unit 2	Unit 3	Unit 4	Unit 5
Materials Fundamentals	Steel & Alloys	Testing & Quality	Manufacturing	Joining & Failure
Wk 1 Aug 24 Atomic Structure & Bonding	Wk 4 Sep 14 Carbon & Alloy Steels	Wk 7 Oct 5 Mechanical Testing	Wk 10 Oct 26 Casting	Wk 14 Nov 23 Welding Processes
Wk 2 Aug 31 Mechanical Properties	Wk 5 Sep 21 Stainless & Cast Irons	Wk 8 Oct 12 NDT Methods	Wk 11 Nov 2 Forming	Wk 15 Nov 30 Weld Metallurgy & Failure
	Wk 6 Sep 28 Nonferrous & Phase Diagrams	Wk 9 Oct 19 Quality Standards	Wk 12 Nov 9 Machining	Wk 16 ← here Dec 7 Corrosion & Wrap-up
			Wk 13 Nov 16 Powder Metallurgy	

Week 17 • FINAL EXAM • December 16, 2026

SLIDE 4 MECH 191 — Course & Unit Roadmap

NOTES

Topics

Resources

Moniz Ch. 31
Corrosion

Stainless Steel PREN — Grade Comparison
Corrosion Protection Guide — NACE / ISO 12944
ASM Vol. 13A — Corrosion Fundamentals

NACE SP0169 — CP Criteria
ISO 8501 — Surface Preparation
NACE MR0175 — H₂S Service

5:00 – 7:00 PM • Same location as class

NOTES

MECH 191 — Metallurgy · Week 16 Notes · fill in blanks & key terms as you go

Try to answer these before we dive in — we'll revisit them with answers at the end of class.

- ## SLIDE 6 Core Questions

Corrosion Types — Uniform & Galvanic

Uniform (General) Corrosion

Mechanism

Anodic and cathodic reactions distributed uniformly across the surface — no localization. The most predictable form of corrosion because the rate is measurable and stable.

Recognition

Even surface loss, uniform scale or rust layer. Corrosion rate measured in mm/year or mils per year (MPY) — allows remaining life calculations.

Design approach

Specify a material with a known corrosion rate for the environment, then add a corrosion allowance to the wall thickness. Monitor with UT thickness measurement at regular intervals throughout service life.

Examples

Rusting of unprotected carbon steel in atmosphere or soil. Acid attack on steel in pickling baths. Uniform thinning of tank shells in process environments.

Galvanic Corrosion

Mechanism

Two dissimilar metals in electrical contact in an electrolyte — potential difference drives current; the less noble metal (anode) corrodes at an accelerated rate while the more noble metal (cathode) is protected.

Area effect — critical

Large cathode + small anode = severe attack; large anode + small cathode = mild.
Zinc on steel works: Zn is the large anode, exposed steel at scratches is a tiny cathode. Never coat the anode only — any break creates a concentrated small-anode attack.

Recognition

Concentrated corrosion on the less noble material, particularly near the junction between the two metals. The more noble material shows little or no attack.

Prevention & Examples

Electrical isolation (non-conducting gaskets, sleeves); select metals close in galvanic series; apply protective coating to the cathode. Examples: zinc anodes on ship hulls; Al fittings on SS pipework; carbon steel fasteners in copper alloy corroding rapidly.

SLIDE 8 Corrosion Types — Uniform & Galvanic

NOTES

Corrosion Types — Pitting, Crevice & Stress Corrosion Cracking

Pitting & Crevice Corrosion

Pitting Corrosion

Local passive film breakdown → small anode surrounded by large passive cathode → pit becomes self-sustaining. Metal ions accumulate inside the pit, attracting Cl^- , dropping pH, preventing repassivation. Recognition: discrete pits on otherwise intact surface — depth concealed by thin corrosion product.

$\text{PREN} = \%Cr + 3.3 \times \%Mo + 16 \times \%N \rightarrow \text{higher} = \text{better pitting resistance}$

304 (PREN ~19) • 316 (PREN ~24) • Duplex 2205 (PREN ~35-40)

Crevice Corrosion

Stagnant electrolyte in a tight gap (0.025–0.1 mm) depletes oxygen → interior becomes anodic → metal ions accumulate, pH drops, Cl^- migrates in → aggressive chemistry prevents repassivation. Recognition: damage concentrated under gaskets, lap joints, bolt heads, and deposits.

Prevention: butt joints over lap joints; solid/PTFE-filled gaskets; continuous rather than intermittent welds; higher PREN alloys; ensure drainage.

Stress Corrosion Cracking (SCC)

Mechanism

Tensile stress (residual or applied) + specific corrosive environment = cracking at stresses far below yield strength. Involves anodic dissolution at crack tip and hydrogen embrittlement. Crack propagates stably until the remaining cross-section fractures by overload.

Critical material-environment pairs

Austenitic SS + chlorides. High-strength steel + H₂S (sulfide stress cracking — NACE MR0175). Brass + ammonia (season cracking). Aluminum alloys + chlorides. Titanium + methanol or red fuming nitric acid.

Recognition

Branching, intergranular or transgranular crack network. Minimal plastic deformation. Brittle fracture surface with no beach marks — this is the key distinction from fatigue. Branching is the most characteristic macroscopic feature visible in the field.

Prevention & Detection

Stress relief (PWHT), shot peening for compressive RS. Select resistant alloys (duplex SS for chlorides, low-alloy steel + PWHT for H₂S). Modify environment (remove Cl⁻ or O₂; inhibitors; maintain pH >7). Detection: PT, ACFM through coatings. UT for subsurface.

SLIDE 9 Corrosion Types — Pitting, Crevice & Stress Corrosion Cracki

NOTES

Corrosion Protection — Coatings & Cathodic Protection

Protective Coatings

Organic paint systems

Three-coat system: zinc-rich primer (sacrificial — zinc corrodes to protect steel even when scratched) + intermediate barrier coat + UV/abrasion-resistant topcoat. Surface preparation is the single most important factor — blasting to Sa 2.5 per ISO 8501. Holiday testing detects pinholes before commissioning.

Hot-dip galvanizing

Steel immersed in molten zinc at $\sim 450^{\circ}\text{C}$ — zinc metallurgically bonds through a series of Fe-Zn alloy layers. Provides both barrier and sacrificial protection. Zinc corrodes preferentially at scratches and cut edges. Service life 20–50 years in mild environments.

Thermal spray

Zinc or aluminum powder/wire melted and propelled onto the surface — applied to large structures in situ (bridges, offshore structures). Thicker than electroplating; can be applied without immersion.

Anodizing & conversion coatings

Anodizing thickens the oxide on aluminum electrochemically — improves corrosion and wear resistance. Phosphating provides paint-adhesion base for automotive steel. Chromate conversion being phased out (toxicity); trivalent alternatives increasingly specified.

Cathodic Protection & Inhibitors

Cathodic protection (CP)

Makes the structure the cathode of an electrochemical cell — corrosion only occurs at anodes.

Sacrificial anode CP

Zn, Mg, or Al anode connected to the structure — the anode corrodes preferentially. Passive, no power required, self-regulating. Ship hulls, buried pipelines, tank floors. Anodes must be inspected and replaced periodically.

Impressed current CP (ICCP)

DC power source drives current from inert anode (graphite, MMO) onto the structure. Controllable and suitable for large installations. Buried/submerged pipelines, storage tanks, reinforced concrete. Requires monitoring — power supply failure leaves structure unprotected.

CP criterion (NACE SP0169): -0.85 V vs. Cu/CuSO₄ reference electrode

Inhibitors

Anodic (form anode film — risk if under-dosed) · Cathodic (reduce cathodic rate — safer under-dosed) · Mixed (act on both — cooling water, pipelines, preservation fluids).

SLIDE 10 Corrosion Protection — Coatings & Cathodic Protection

NOTES

Design, Material Selection & Corrosion Inspection

Design & Material Selection

Material selection is the most permanent defense — an inherently resistant alloy requires no maintenance and won't fail if the protection system is damaged.

Joint design

Butt joints preferred over lap joints — lap joints create crevices. Continuous welds instead of intermittent welds seal gaps. Full-penetration seal welds at tube-to-tubesheet joints eliminate the crevice between tube OD and bore.

Drainage & surface finish

Slope surfaces so water drains — horizontal surfaces that pool water corrode faster. Smooth surfaces are more corrosion resistant than rough (rough surfaces trap moisture and contamination). Smooth machined keyways + polished fillets reduce both corrosion and fatigue initiation

Dissimilar metal isolation

Electrically isolate dissimilar metals with non-conducting gaskets, sleeves, and washers. If isolation is impractical, select metals close together in the galvanic series or ensure the less noble material has a much larger area than the more noble

Residual stress control

Tensile residual stress from welding or machining accelerates SCC. PWHT for stress relief. Shot peening introduces compressive residual stress at the surface — a common SCC and fatigue prevention measure for welded joints and keyways.

Corrosion Inspection & Monitoring

Visual inspection

First step — rust staining, discoloration, paint blistering, pitting, deposits, and scaling. Requires adequate lighting, access, and mirrors or borescopes for confined spaces.

UT thickness measurement

Primary method for measuring remaining wall thickness in corroded piping, vessels, and structural members — from one side without internal access. Results compared to minimum acceptable thickness (fitness for service).

Corrosion Under Insulation (CUI)

Corrosion hidden beneath insulation — cannot be seen without stripping. Pulsed eddy current (PEC) and radiographic profile techniques screen through insulation without removal. A significant inspection challenge in process plant.

Monitoring — rate measurement

ER probes and LPR probes: real-time corrosion rate in process pipelines and vessels. Corrosion coupons: pre-weighted metal samples exposed for a defined period — direct weight loss measurement. Used to verify inhibitor programs and detect changes in process chemistry.

SLIDE 11 *Design, Material Selection & Corrosion Inspection*

NOTES

Course Wrap-Up — How the Five Units Connect

These are not five separate subjects. They form a single chain — each unit explains why the next one matters.

Unit 1 — Materials Fundamentals Aug 24 – Sep 7	Atomic bonding explains conductivity and electrochemical corrosion. Crystal structure determines slip systems, deformation behavior, and SCC crack paths. Grain boundaries are sites of sensitization, intergranular corrosion, and hydrogen trapping.
Unit 2 — Steel & Alloys Sep 14 – Sep 28	Carbon equivalent drives preheat requirement and determines HIC susceptibility. Stainless grade determines pitting (PREN) and SCC resistance. Heat treatment condition sets HAZ hardenability — the microstructure that either resists or fails in service.
Unit 3 — Testing, Inspection & Quality Oct 5 – Oct 19	NDT methods detect the defects that Unit 2's microstructure makes and Unit 4's processes create. Hardness testing verifies heat treatment. Inspection documentation creates the traceability that makes a failure investigation possible.
Unit 4 — Manufacturing Processes Oct 26 – Nov 16	Casting defects provide corrosion and fatigue initiation sites. Forming introduces residual stress that drives SCC. Machining quality determines surface finish and residual stress state — the foundation for fatigue life in service.
Unit 5 — Joining, Failure & Corrosion Nov 23 – Dec 7	The weld HAZ is a microstructure gradient each zone with different corrosion resistance. Residual weld stress drives SCC. Failure analysis connects all previous units — explaining why a component failed and feeding that knowledge back into design, material selection, and inspection.

SLIDE 12 *Course Wrap-Up — How the Five Units Connect*

NOTES

Corrosion type identification in the field — what the technician sees and what it means.

Inspection reveals three overlapping corrosion mechanisms: (1) Galvanic — carbon steel tubesheet is anodic to stainless tubes; unfavorable area ratio (small anode/large cathode) drives accelerated attack at the tubesheet. (2) Crevice — the tight gap between tube OD and tubesheet bore in stagnant seawater depletes O_2 , drops pH, concentrates Cl^- — classic crevice geometry. (3) Sensitization — if standard 304 (not 304L) was used without interpass temperature control, chromium carbide precipitation in the HAZ depletes Cr below the passivity threshold. Action: specify 304L or duplex 2205 for seawater; specify full-strength seal welds to eliminate the crevice; apply sacrificial zinc anodes to the carbon steel tubesheet; add to failure investigation as a multi-mechanism case study.

VT finds tight, branching cracks running from weld toes. No beach marks on the fracture surface (distinguishing SCC from fatigue). No plastic deformation. Metallographic sections show transgranular cracking with fine branching — the textbook appearance of chloride SCC in austenitic stainless. Root cause: tensile weld residual stress + chloride from insulation + elevated temperature. Action: PWHT for stress relief (not always possible on austenitic SS — sensitization risk); shot peen weld toes; review insulation specification (chloride-free insulation required per NACE for SS service); consider upgrade to duplex stainless or super-austenitic for remaining vessels in the same service.

A newly coated pipeline section is holiday tested before backfilling using a high-voltage spark tester. A pinhole is found at a fitting where the coating was not fully sealed. The pinhole is repaired and retested. Without the repair, the pinhole under CP would have created a small anode/large cathodically protected cathode situation — concentrating the full galvanic driving force at a single point and producing rapid localized attack. Action: holiday testing is a mandatory step before burying or commissioning any cathodically protected coated structure; document all findings and repairs; retest after each repair until the coating is holiday-free.

NOTES

MECH 191 — Metallurgy · Week 16 Notes · fill in blanks & key terms as you go

Core Question Answers

What is corrosion and why is it thermodynamically driven?

Corrosion is the degradation of a metal through chemical or electrochemical reaction with its environment. It is thermodynamically driven because most metals exist naturally as ores (lower energy state) — refining requires energy input, and corrosion is the metal releasing that energy and returning toward its natural state. Electrochemical corrosion requires four elements simultaneously: anode, cathode, electrolyte, and metallic path. Remove any one element and corrosion stops.

What are the main types of corrosion and how is each recognized?

Uniform: even surface loss, predictable rate. Galvanic: concentrated attack on the less noble metal at a dissimilar metal junction. Pitting: discrete pits on otherwise intact surface — depth concealed by corrosion product; driven by local passive film breakdown and Cl^- concentration; PREN quantifies resistance. Crevice: damage under gaskets, lap joints, deposits — stagnant O_2 -depleted chemistry. SCC: branching, intergranular/transgranular cracks with no beach marks, no plastic deformation, driven by tensile stress + specific environment.

What protection methods are available and how do you choose between them?

Coatings (barrier and sacrificial) + surface prep is #1 factor. Hot-dip galvanizing for structural steel: barrier + sacrificial, 20–50 yr life. Cathodic protection: sacrificial anodes (passive, ships/pipelines) or ICCP (controllable, large structures). Inhibitors: anodic (risk if under-dosed), cathodic (safer). Material selection: most permanent — match PREN, CE, and alloy to the specific environment. Design: eliminate crevices, ensure drainage, isolate dissimilar metals, control residual stress. Most effective systems combine multiple methods.

How does material selection and design influence corrosion risk?

Material selection must match the specific corrosion mechanism — 304 SS resists uniform attack but fails in Cl⁻ pitting; copper is excellent in freshwater but fails in ammonia. PREN guides stainless selection for chloride service. CE and alloy composition determine weldability and HAZ hardenability. Design decisions: butt joints over lap joints (no crevice); sloped surfaces for drainage; non-conducting isolation for dissimilar metals; smooth surface finish; PWHT and shot peening to control residual stress and prevent SCC.

SLIDE 14 *Core Question Answers*

NOTES

Week 16 Resources

Corrosion and Its Prevention — primary reading

Service environment guide for stainless alloy selection

Coating selection, surface preparation, and cathodic protection

Standard for cathodic protection of buried pipelines — CP criteria

Material requirements for H₂S service (sour service)

Primary reference for corrosion mechanisms and materials selection

Unit 1 — Atomic structure, bonding, mechanical properties

Unit 2 — Steel families, heat treatment, phase diagrams

Unit 3 — NDT methods, mechanical testing, quality standards

Unit 4 — Casting, forming, machining, powder metallurgy

Unit 5 — Welding, weld metallurgy, fracture, corrosion

SLIDE 16 *Resources & Final Exam*